

BIM - Building Information Modeling

- definition: generating and managing building data
 - during design, construction, building or asset life cycle
- multiple stakeholders (AEC, architecture, engineering, construction)
- modeling is the process, not model
- N-dimensional
 - 3D: x, y, z
 - 4D: time
 - 5D: cost

BIM vs CAD

- | | |
|---------------------------------|------------------------|
| - features (wall, window, door) | - point, line, polygon |
| - elements are associated | - no association |
| - change once | - change every layer |
| e.g. material, cost | - limited information |

Stakeholders

- AEC, supplier
- Non-BIM users: CEO, investor, government, media

Life cycle of Building Project

- programming → conceptual design → detailed design
- analysis → documentation → fabrication → construction
- operation and maintenance

Goal of BIM

- sustainable infrastructure
- operation and management of buildings

BIM vs GIS

- | | |
|----------------------------------|----------------------|
| - buildings and facilities | - regions, countries |
| - mm level | - m level |
| - focus on facility lifecycle | - wide ranging |
| - focus on construction material | - generic geometry |
| - no network | - models network |
| - local coordinate system | - global |

Why integrate BIM → GIS?

- compatibility
- interoperability
- reuse

BIM in Hong Kong examples

- CIC
- HKU, UST, PolyU (CIC)
- GeoBIM

Functions of BIM

- ① Visualization
- ② Collaboration
 - improve efficiency
 - communication, common platform
- ③ Simulation
 - emergency, wind, light, fire
 - 3D, 4D (time), 5D (cost)
 - accurate and real time
- ④ Optimization
 - geometry and spatial analysis
- ⑤ Cartography
 - 3D to 2D

Roles of BIM in Construction Stages

- Main Objectives
 - time
 - cost
 - performance (quality)
- Typical Life cycle
 - initialing
 - planning
 - construction
 - monitoring and control
 - closing

① Modeler

- import maps, documents into BIM

② Engineer

- generate BIM model, work with designers

③ Project Manager

- coordinate technical tasks
- BIM application platform
- standardized workflow

④ Director

- integrate different resources
- make plans for different departments

Standards in Hong Kong - departments

- CIC . HK Housing Authority

International : ISO 19650

BIM Software examples

- Revit . 3Dmax . Bentley - micro station

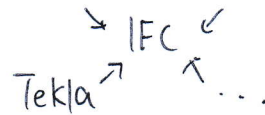
Open BIM

- data sharing
- collaboration across platforms and stakeholders
- including Industry Foundation Classes (standards)

IFC

- global standard
- BIM data exchange
- maintained by building SMART
- advantage : Interoperability

- archiCAD Revit



- use IFC files
- current version : IFC 4.3

- Structure of IFC files

- Header : - general information about the model
 - IFC version and software used
- Body : - geometry and attributes of the building
- advantage :
 - particular attributes are stored only once
 - reduce file size

- IFC file formats

- official : STEP . XML . ZIP , Turtle . RDF
- candidate : JSON
- experimental / unsupported : SQLite
- most widely used : STEP
 - extension : .ifc - adv : most compact with text
 - text ✓ , index X
 - size : 100%
 - MIME type : application / x-step

IFC File Format - STEP

- machine readable modelling language (Express)
- reference sub models

Express

- multiple inheritance

IFC File Format - XML

- application / xml
- size : 113 %
- enhanced readability
- extension : .ifcXML
- adv :
 - platform independent
 - any language or OS

IFC File Format - ZIP

- application / zip
- extension : .ifcZIP
- size : 17 %
- no text

Other : JSON : ^{size} 148 % HDF (candidate)
Turtle : 1372 %

IFC data structure - object model → 2 entities

- hierarchy of object instances
- all^{ID} objects derive from IfcRoot
- every object has a GUID (UUID)
- non-root objects
 - no identity
 - instance exist by reference
- objects by types :
 - construction objects (eg. walls, IfcWall)
 - geometric objects
 - structural objects

IfcRoot

- three abstract concepts
 - object definition : Ifc...
 - relationships : Ifc...
 - property sets : IfcPropertyDefinition

IfcObject

- 6 categories
- actor, control, group, product, process, resource

IfcProduct

- 3 types of elements
- spatial, physical, structural

IfcRelationship

- 5 relationships
- IfcRel...
- decomposes, assigns, connects, associates, defines

IFC data

- overall, IFC standardizes ..
- objects, relationship
- people, processes
- attributes (material, color)
- abstract concepts

Geometric representation of IFC objects

- 3 ways
- extrusions
- solid body representation using a sweep
- using B-reps (boundary representation)

Components of GIS

- people, procedures
- software, hardware
- data

3D GIS

- incorporating Z-value into mapping
- elevation / other variables as Z-values
- 3 types of ground objects
 - building, infrastructure, terrain

City GML (3.0)

- xml based storage
- storage and exchange of virtual 3D city models
- cost effective and sustainable
 - reuse same data in different application fields
- international standard by OGC

General characteristics of City GML

① Modularization

- split into core module and thematic extension modules
- 13 extension modules

② Multi scale modeling - 5 LOD

- LOD 0 to 4
- efficient visualization
- combine 2 CityGML datasets with different LOD

③ coherent semantical-geometrical modeling

- 2 hierarchies: semantical, geometrical
- 2 levels: semantic, spatial

General characteristics of CityGML

④ Closure surfaces

- objects require closure to compute volume

⑤ Terrain Intersection Curve (TIC)

- solve the problem where 3D objects intersect the terrain

⑥ Code list for enumerative attributes

- e.g. habitation → looo, sanitation → lololo
- attributes values → list of discrete values
- Purpose: ensure same code is used for the same notion / concept

⑦ External References

- using external data sets

⑧ City Object Groups

- groups contain objects or groups

⑨ Appearance

- information about the surface appearance
- not just visual (e.g. infrared radiation, noise pollution)

⑩ Prototypic objects

- e.g. trees, traffic lights, furniture

Softwares

- Esri: City Engine, ArcScene, ArcGIS Pro
- Autodesk: LandXplorer
- Aristotles Viewer

CityGML Summary

- define geographic objects in the city
- geometry, topology, semantics, appearance

Advantages of City GML

- young specification
- solve the problems of city models
- focus on
 - model storage
 - management
 - exchange
- reusability

City GML Summary

- describe physical and semantic properties
- city elements: buildings, roads, rivers, vegetation

Strengths:

- data exchange format for 3D GIS
- open standard - cross industry adoption
- different LOD
- object oriented
- semantic models
- facilities management

Limitations

- XML: no spatial index
 - not optimized for storage
- complex data model
- difficult to stream
- need conversion to web: pdf, 3js, geojson
- not heavy enough for true BIM
 - main focus: community and city scale

IFC vs City GML

- | | |
|--------------------------|----------------------------|
| - IFC is BIM | - City GML is GIS |
| - modeling new buildings | - modeling existing data |
| - focus on objects | - focus on area |
| - building scale | - city and worldwide scale |

Georeferencing

- | | |
|---|---|
| - bottom-up | - top-down |
| - idea → design → real world | - real world → surveying → 2D/3D model |
| - relative position | - absolute position |
| - local project coordinate system (PCS) | - regional/global coordinate reference system (CRS) |

Modeling paradigm

- | | |
|-----------------------------------|------------------------------------|
| - bottom-up | - top-down |
| - aim: correct repr. of buildings | - aim: correct repr. of real world |
| - construction elements | - only the surface |

Geometry Representation

- | | |
|--------------------------|--|
| - parametric, sweep, CSG | - B-Rep |
| - logical operator | - aggregation of all surfaces surrounding the volume |

Standards

- | | |
|------------------------------------|------------------|
| - buildingSMART | - OGC - CityGML |
| - IFC | - based on xml |
| - STEP / ifcxml | - has appearance |
| - no representation for appearance | |

IFC and City GML conversion

- matching entities
- n City GML entities may represent one IFC element (or m elements)
- mostly conversion of geometry
- downgrading IFC to low LOD City GML
- method: development of an ADE
 - near lossless conversion
 - storing only the interested data
 - develop from scratch
 - OR
 - modify / extending existing ADE

Technical Challenges

1. Accommodate with massive BIM data
 - lower LOD
 - instantiate drawing for visualization
 - lightweight application
2. How to match multi-source data
 - data fusion
 - unify BIM model
 - terrain and other multi-source data
 - TIN, water surface, photography model
3. Spatial Analysis
 - spatial relationship (eg. contain, intersect)
 - spatial query application
 - BIM and TIN spatial computing
4. Multiple devices and applications
 - desktop app, mobile app, webpage

City Information Model (CIM)

- holistic approach to city planning
- from build level to city level
- Purpose: urban planning, management, smart computing
- include buildings, roads, public spaces

What is CIM

- integration of BIM, GIS, IoT
- decision making model
- model management system (BIM + GIS)

Why combine BIM and GIS?

- both outdoor and indoor information
- surface model: GIS
- internal model: BIM
- both are semantic enhanced
- key: data fusion of IFC and City GML

Integrating IFC and City GML

- Application Domain Extension (ADE)
- Unified Building Model (UBM)
- conversion of IFC to City GML
- IFC ~ 900 class, most cannot map to City GML
 - 60 ~ 70 classes similar to City GML
- City GML: 1 core module, 11 thematic modules
- IFG: IFC + GML → exchange information between GIS and CAD

Sub-models of CIM

- building model
- transportation model
- city furniture model
- mechanical, electrical, plumbing (MEP) model
- waterbody model

Objective of CIM

- building a city info visualization platform
- decision-making
- urban operation management platform
- city service platform

Framework

- Four-tier Architecture

- infrastructure layer :
 - basic computing resources
 - storage resources
 - network resources
 - data layer :
 - multi-source data
 - efficient storage management
 - platform layer :
 - big data collection
 - service resource resource centers
 - development centers
 - application layer :
 - establish different systems
 - implement supervision
 - multi-level support
- 1,2,3 tier architecture : client, logic, data

CIM Solution

- BIM + GIS + IoT
- Real time operation and management
- new technologies
 - multi-source spatial data fusion
 - urban neuro network
 - urban computing
 - urban sensing
 - urban intelligence
 - city modules

3D data integration

- coordinate transformation
- Spatial calculations and operations for terrain, oblique imagery

S3M: Spatial 3D Model

- open source
- format conversion between S3M and 3D tiles

Challenges

- dynamic, direction, diversity

20/80 Rule

- most nodes: few connections, vice versa
- 80% effect caused by 20%

Head/tail division

- Split 20/80 by mean value

Network structure - relative complexity

- γ index

$$\gamma = \frac{l}{l_{\max}} = \frac{l}{3(n-2)} \quad n: \text{no. of nodes}$$

- ratio of links to max. possible links
- close to 0 : simple network
- close to 1 : better connected network

Network analysis

- degree : no. of connections
- betweenness : no. of shortest paths that pass through the node
- closeness : sum of length of shortest paths from node to other nodes

Modeling Space : space syntax

- set of techniques for spatial analysis
- Axial map
 - least no. of largest lines

Power law distribution

- more small geographic objects than larger ones

Multi-layer network for CIM

- network A : building elements
- network B : space syntax

weighted network

W : edge length, 0 = self, ∞ = no node

D : shortest distance matrix

Complexity of map and space

- visual complexity : visual process
- intellectual complexity : cognitive process

Law of simplicity

- when presented with ambiguous or complex objects, brain makes it appear simple

Law of proximity

- group close things together

Law of continuity

closure, area, symmetry

Livingness

$$L = S \times H$$

- substructures
- hierarchy

Visualization summary

- buildings and cities are more organ-like, instead of machine-like
- nature of space lies on livingness
- Goodness of urban design is a fact rather than opinion
- building/city $\rightarrow$ coherent whole
 - $\rightarrow$ sense of beauty

Digital Twin

- digital repr. of physical object, person, or process
- also integration of BIM and 3D GIS (other than CIM)
- BIM + GIS + IoT
- also a solution
- Objective :
 - simulate real situations
 - ultimately make better decisions
- Challenges :
 - different data, systems, sets of information
 - development and delivery of smart cities
 - complexity and dynamic interaction
 - large number of stakeholders
 - without tangible operational model
 - $\rightarrow$ combine cross-sector data and info
- 2 Aspects :
 - model framework is spatially and temporarily accurate
 - dynamic properties $\rightarrow$ adapt to change

Digital twin in urban scale

- goal: transform how city is planned, built, managed.
- make sustainable urban environment
- mechanism must be responsive, interactive

Digital twin in practice

- (SDI) platform in Hong Kong
- 3D model of HKUST

Challenges of digital twin

- user focus: usability, usefulness
- integration, interoperability
- openness
- regulation: security, privacy, decision with AI

Spatial Big Data analysis

- urban big data computing
- multi-plex network analysis

difficulties

- large scale, highly dynamic
- cloud computing: does not support ST data

Urban computing framework

- Sensing → computing → decision

- example: LBS

- key: design and develop additional service
- privacy problems
- privacy protection: geodetic solution

$$n' = \frac{\text{floor}(n * r + 0.5)}{r}$$

- randomization

3D Analysis

- visual analysis: visible points
- visibility area
- shadow, shadow rate
- sun shine, city skyline

Why urban informatics

- uncertainty in urban big area
- urban discovery, prediction, explanation
- complexity science

Core technologies

- AI based urban remote sensing
- 3-D mobile mapping
- spatial big data analytics and platform

objectives

- relations between cyber space and urban environment
→ quantify relation from complexity science

Challenges

- human interaction in hybrid space
- integration repr. of hybrid human interaction
- death of distance in hybrid space

Complex systems theory

- require geography domain knowledge
- notation and influences of multi-dimensional space
- Network as part of geography framework, not the whole
- 'interaction' in urban and geography context